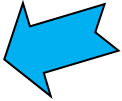


Module 6. Cluster Analysis and Selected Methods

- ❑ Cluster Analysis: Basic Concepts
- ❑ Partitioning Methods
- ❑ Hierarchical Methods
- ❑ Density-based and Grid-based Methods 
- ❑ Advanced Methods
- ❑ Evaluation of Clustering
- ❑ Summary

Density-Based Clustering Methods

- ❑ Clustering based on density (a local cluster criterion), such as density-connected points
- ❑ Major features:
 - ❑ Discover clusters of arbitrary shape
 - ❑ Handle noise
 - ❑ One scan (only examine the local region to justify density)
 - ❑ Need density parameters as termination condition
- ❑ Several interesting studies:
 - ❑ DBSCAN: Ester, et al. (KDD'96)
 - ❑ OPTICS: Ankerst, et al (SIGMOD'99)
 - ❑ DENCLUE: Hinneburg & D. Keim (KDD'98)
 - ❑ CLIQUE: Agrawal, et al. (SIGMOD'98) (also, grid-based)